TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Water Repellency and Water Absorption

Two opposing faces of water: keeping rain off the surface, and drinking liquid up into the fabric — spray rating, wicking and immersion absorption measured side by side.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 9 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine (a) the surface water repellency of a fabric by the spray test (ISO 4920, equivalent in principle to AATCC 22), expressed as a spray rating, (b) its absorbency by drop sinking, vertical wicking height and rate, and immersion water absorption percentage, and to interpret the combination of results for end uses that need shedding, absorbing, or both behaviours in different layers.

1.1 Repellency and absorbency are different properties

Students routinely confuse three water behaviours that this experiment separates. **Water repellency** is a surface property: drops bead, sit on the face and roll or shake off without wetting it. **Water absorption (absorbency)** is a bulk property: liquid is taken up into fibres and into the capillary spaces between them, so the fabric gains weight and transports liquid along itself. **Waterproofness** is a third property again — resistance to penetration under pressure, measured by a hydrostatic head test — and a fabric can be highly repellent yet not waterproof, because a repellent finish stops gentle wetting but forced water still passes through the pores. This sheet covers the first two; hydrostatic head belongs to coated and protective-fabric testing.

Both behaviours grow out of the same physics: surface tension and contact angle. On a low-energy surface — a fluorocarbon or silicone finish, or a naturally waxy or tightly structured face — water meets the solid at a high contact angle, beads up, and the capillary pressure in the fabric pores opposes entry. On a high-energy, hydrophilic surface — clean cotton, viscose, scoured wool — the contact angle is low, water spreads, and the same capillaries pull liquid in and along with force. Finishing can therefore move a single base fabric in either direction: a durable water-repellent (DWR) finish raises repellency without sealing the pores (the fabric stays breathable), while scouring, removal of natural waxes, and hydrophilic softeners raise absorbency. A rain jacket wants the face repellent and the pores open; a towel wants the opposite in every fibre; a sportswear base layer wants fast wicking along the fabric with little retention — three different specifications, all measurable on this bench.

1.2 The spray test: ISO 4920 / AATCC 22

In the spray test a measured volume of water — 250 mL — is poured through a standard funnel and spray head so that it falls as a controlled shower onto a specimen mounted at 45°, the centre of the specimen 150 mm below the spray nozzle. The specimen is then tapped sharply to shake off loose drops and its wetted face is compared, under standard viewing, with the photographic spray-rating chart. The rating is a description of the wetting pattern, not a percentage:

SPRAY RATING	ISO GRADE	APPEARANCE OF THE SPRAYED FACE
100	ISO 5	No sticking or wetting of the upper surface.
90	ISO 4	Slight random sticking or wetting of the upper surface.
80	ISO 3	Wetting of the upper surface at the spray points.
70	ISO 2	Partial wetting of the whole upper surface beyond the spray points.
50	ISO 1	Complete wetting of the whole upper surface.
0	—	Complete wetting of the whole upper and lower surfaces (strike-through).

Two cautions belong to every report. First, the scale is ordinal and non-linear: the difference between 100 and 90 is not the same quantity of water as the difference between 70 and 50, and ratings must never be averaged as if they were measurements — report the individual ratings, and where duplicates disagree, the test is repeated. Second, the spray test rates only the face under a gentle shower; it predicts rain-shedding and shower resistance, not performance under pressure, immersion or prolonged rain, and a rating falls as a DWR finish is washed out — which is itself a useful durability check when specimens are rated before and after laundering.

1.3 Absorbency tests

Three complementary observations cover absorbency. In the **drop test**, a drop of water placed on the conditioned face is timed until its bright specular reflection disappears — the sinking time. A truly absorbent fabric takes the drop in a second or less; a repellent fabric holds it beaded indefinitely, and the test is stopped and recorded as non-absorbent. In the **vertical wicking test**, a strip is suspended with its lower end dipping into water (often lightly dyed so the front is visible) and the height reached by the rising front is read after set times: capillary rise balances capillary pull against gravity, so height and rate together describe how fast a fabric will move sweat or spilt liquid away from a point. Warp and weft strips wick differently because the yarns and their capillary channels differ. In the **immersion test**, a weighed dry specimen is immersed, removed, drained under standard conditions and reweighed:

$$\text{Water absorption (\%)} = (\text{wet mass} - \text{dry mass}) / \text{dry mass} \times 100$$

$$\text{Wicking rate (mm/min)} = \text{wicking height (mm)} / \text{time (min)}$$

Immersion absorption counts water inside fibres plus water held between them, so bulky, lofty structures score high even with moderately hydrophilic fibres, while a dense fabric of very hydrophilic fibre can score lower. All absorbency testing is done after conditioning at 65% RH and 20 °C, and — critically — on fabric in the state to be judged: residual softener, silicone or an unscoured loom-state wax can make an absorbent fibre test as repellent for the first washes of its life.

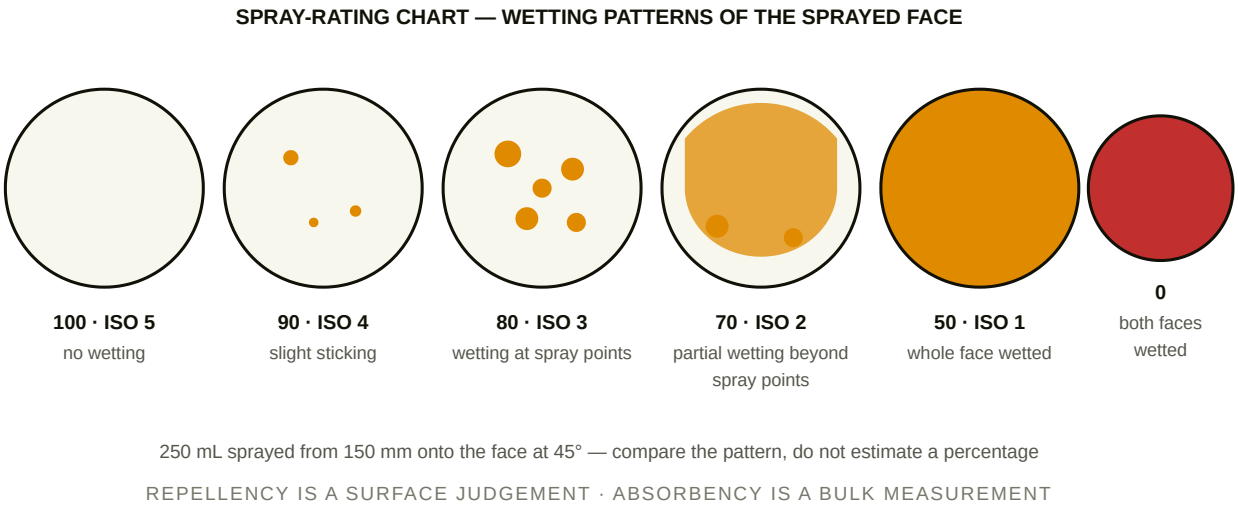


Figure 1 — The spray-rating reference patterns. From left: rating 100 (ISO 5), face completely dry; 90, slight random sticking; 80, wetting confined to the spray points; 70, wetting spreading beyond them; 50 (ISO 1), the whole face wetted; 0, wetting through to the lower surface.

2. Apparatus and procedure

- **Apparatus:** spray tester — funnel, standard spray nozzle, specimen hoop and stand giving a 150 mm drop to a specimen held at 45°; spray-rating chart; measuring cylinder (250 mL); wicking frame with ruler, reservoir and lightly dyed water; stopwatch; balance (0.01 g); immersion tray and draining grid; conditioned specimens.
- **Conditioning:** condition all specimens in the standard atmosphere (65% RH, 20 °C) and test there. Use distilled or deionised water at room temperature throughout, so dissolved salts do not alter surface tension between repeats.
- **Spray test:** mount the specimen face-up in the hoop, smooth but not stretched, centre it 150 mm below the nozzle, and pour 250 mL of water steadily through the funnel so the shower falls evenly. Remove the hoop, tap it sharply twice against a hard edge to dislodge loose drops, and compare the

face at once with the rating chart under good light. Record the rating for each of at least three specimens and report them individually.

- **Drop test:** lay a conditioned specimen flat, deliver one drop from a fixed small height at each of several places, and time until the bright reflection disappears. Record the mean sinking time, or record the fabric as non-absorbent if drops remain beaded after the agreed limit.
- **Wicking:** cut warp-way and weft-way strips of fixed width, mark a datum line, suspend them vertically with the datum just at the water surface, start the stopwatch, and read the height of the visible front above the surface after set intervals — 1, 5 and 10 minutes in this exercise. Read the lowest continuous front, ignoring isolated fast threads, and state the rule used.
- **Immersion absorption:** weigh the dry conditioned specimen, immerse it fully for the standard time, remove it, drain it in the standard manner for the standard time, and reweigh at once. Calculate absorption percentage. Repeat on duplicate specimens.

3. Worked calculation

Spray test. Three specimens of a DWR-finished jacket fabric all showed wetting confined to the spray points, with the rest of the face dry after tapping. Rating assigned to each specimen:

$$\text{Spray rating} = 80 \text{ (ISO 3)} \text{ — wetting of the specimen face at the spray points}$$

A rating of 80 is a sound commercial result for a breathable rain-shedding face: the finish is working over the whole surface (a failing finish drops quickly to 70 and below, where wetting spreads freely between spray points), but the fabric is not at the 90–100 level of a new, premium DWR. In a durability study the same fabric would be re-rated after 5, 10 and 20 washes to track finish loss.

Wicking. Heights reached in 10 minutes: warp-way strip 62 mm, weft-way strip 48 mm.

$$\text{Warp wicking rate} = 62 / 10 = 6.2 \text{ mm/min} \quad \cdot \quad \text{Weft wicking rate} = 48 / 10 = 4.8 \text{ mm/min}$$

The warp direction wicks about 29% faster than the weft ((6.2 − 4.8) / 4.8 × 100 ≈ 29%), consistent with straighter, more continuous capillary channels along the warp yarns. Both rates show a genuinely absorbent structure beneath the repellent face — the classic profile of a face-finished fabric whose interior remains hydrophilic, so that any water which does enter is spread and held rather than beading inside.

Immersion absorption. Dry conditioned mass = 10.00 g; mass after immersion and standard draining = 16.40 g.

$$\text{Water taken up} = 16.40 - 10.00 = 6.40 \text{ g}$$

$$\text{Absorption} = 6.40 / 10.00 \times 100 = 64.0\%$$

COMBINED INTERPRETATION

Spray rating 80 (ISO 3) on the face, yet 64.0% immersion absorption and steady wicking (6.2 mm/min warp, 4.8 mm/min weft) in the bulk. There is no contradiction: repellency lives in the finished surface, absorbency in the fibres and capillaries beneath it. Judge each claim — shower resistance, towelling, sweat transport — with its own test, never by inference from the other.

4. Precautions

- Verify the spray geometry before every session — 250 mL, 150 mm from nozzle to specimen centre, specimen at 45°. A tilted hoop or a partly blocked nozzle changes the shower and invents a rating difference.
- Rate the specimen immediately after the two standard taps, under the same lighting, against the chart — not from memory. Delay lets the pattern dry at the edges; extra shaking removes water the standard intends to keep.
- Report spray ratings individually and never as a decimal average. The scale is a set of defined patterns, not a percentage scale.
- Handle specimens by the edges and test the face specified. Finger grease locally destroys repellency and creates a wet spot that belongs to the operator, not the finish.
- In wicking, keep the strip truly vertical, the datum exactly at the water surface, and the reservoir level constant; read the continuous front by the rule stated in the report.
- In immersion, drain for exactly the standard time and manner and weigh promptly — a minute of extra draining or evaporation moves the absorption figure by more than the difference between many commercial fabrics.

Practice (viva preparation)

1. Distinguish water repellency, water absorption and waterproofness.
2. State the spray-test conditions (volume, distance, angle) and what a rating of 80 means.
3. Why must spray ratings not be averaged?
4. A strip wicks 62 mm in 10 minutes warp-way and 48 mm weft-way. Find both rates and comment.
5. A specimen weighs 10.00 g dry and 16.40 g after immersion and draining. Find its absorption percentage.
6. How can one fabric be both repellent (rating 80) and absorbent (64%) at the same time?
7. A DWR fabric rates 90 new and 50 after twenty washes. What has happened and what would you report?

Answer key

1. Repellency is surface behaviour — drops bead and do not wet the face, due to a low-energy surface and high contact angle. Absorption is bulk uptake — liquid enters fibres and capillaries, raising mass and travelling by wicking. Waterproofness is resistance to penetration under pressure (hydrostatic head). A fabric may show any combination: repellent but not waterproof is the normal case for a breathable DWR jacket fabric.
2. 250 mL of water is sprayed through the standard nozzle onto the face of a specimen mounted at 45°, with the specimen centre 150 mm below the nozzle; loose drops are shaken off by two sharp taps and the pattern is matched to the chart. Rating 80 (ISO 3) means wetting of the upper surface at the spray points only — the finish is holding over the general surface but wets where the shower strikes hardest.
3. The chart grades are defined wetting patterns on an ordinal scale with unequal steps in actual water uptake. Averaging 90 and 70 into "80" invents a pattern that may never have occurred and hides disagreement between specimens, which is itself diagnostic of uneven finishing. Individual ratings are reported.
4. Warp rate = 62/10 = 6.2 mm/min; weft rate = 48/10 = 4.8 mm/min. Warp wicks faster, as expected from the straighter, denser warp yarn system offering more continuous capillaries; both rates indicate an absorbent interior structure.
5. Absorption = (16.40 − 10.00) / 10.00 × 100 = 6.40 / 10.00 × 100 = 64.0%.
6. Because the two tests interrogate different layers. The spray test sees only the finished face, whose low-energy treatment makes drops bead. The immersion test forces water past that face and measures the hydrophilic fibres and capillary spaces within, which take up 64% of the dry mass. Face finishes change surface energy without sealing or de-absorbing the bulk.
7. The durable water-repellent finish has been progressively removed or degraded by laundering — detergent action, abrasion and loss of the low-energy layer — so the face now wets completely (rating 50, ISO 1). Report both ratings with the wash count and method, state the durability failure against the buyer's after-wash requirement, and note that re-proofing or a more durable finish system is required; do not report only the new-fabric figure.
